

Shashank Kirtania

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EDUCATION

University of Michigan, Ann Arbor

Ph.D. in Computer Science

2026 - Present

Advised by: Prof. Gabriel Poesia

Thapar Institute of Engineering and Technology

BTech. in Computer Science

2019 - 2023

EXPERIENCE

Microsoft Research, India

Research Fellow (Predoctoral)

Projects: Verified Code Generation

Sep. 2025 – May 2026

Bengaluru

Microsoft, India

Research Fellow (Predoctoral) with the PROSE group

Projects: StackFeed, MetaReflection, Logic-LM++

Sep. 2023 – Sep. 2025

Remote / Bengaluru

Wadhvani AI

Machine Learning Research Intern

Project: Oral Reading Fluency for Indic languages

Jan. 2023 - July 2023

Remote / Bengaluru

Google Summer of Code, PyMC

Open Source Contributor

Project: APIs for probabilistic ML models.

May 2022 - Sep 2022

Remote

PUBLICATIONS AND PREPRINTS

C = CONFERENCE, W = WORKSHOP, J = JOURNAL, P = PREPRINT

P.1 ***“Code is cheap, show me the specification”: Training LLMs for end-to-end Verified Code Generation***
Swayam Singh*, Ramneet Singh*, **Shashank Kirtania***, Tianyu Cheng, Aditya Kanade, Shan Lu, Nagarajan Natarajan
Under review [2026]

C.1 ***Improving Language Agents through BREW: Bootstrapping expeRientially-learned Environmental knoWledge*** 📄
Shashank Kirtania, Param Biyani, Priyanshu Gupta, Yasharth Bajpai, Roshni Iyer, Sumit Gulwani, Gustavo Soares
Conference on Language Modeling 2026 [COLM'26]
Also presented at Multi-Turn Interactions in LLMs at NeurIPS 2025 [MTI @ NeurIPS'25]

W.1 ***IndiMathBench: Autoformalizing Mathematical Reasoning Problems with a Human Touch*** 📄
Param Biyani, **Shashank Kirtania**, Yasharth Bajpai, Sumit Gulwani, Ashish Tiwari
Formal Methods in AI Workshop at AAAI 2026 [FMAI @ AAAI'26]

C.2 ***STACKFEED: Structured Textual Actor-Critic Knowledge Base Editing with Feedback*** 📄
Shashank Kirtania*, Naman Gupta*, Priyanshu Gupta, Krishna Kariya, Sumit Gulwani, Arun Iyer, Suresh Parthasarathy, Arjun Radhakrishna, Sriram K. Rajamani, Gustavo Soares
Empirical Methods in Natural Language Processing 2025 [EMNLP'25]

W.2 ***Activation Steering in Neural Theorem Provers*** 📄
Shashank Kirtania, Arun Iyer
Formal Methods in the Wild Workshop at ICLR 2025 [FMWild @ ICLR'25]

C.3 ***MetaReflection: Learning Instructions for Language Agents using Past Self-Reflections*** 📄
Priyanshu Gupta*, **Shashank Kirtania***, Ananya Singha*, Sumit Gulwani, Arjun Radhakrishna, Sherry Shi, Gustavo Soares
Empirical Methods in Natural Language Processing 2024 [EMNLP'24]

W.3 *Logic-LM++: Multi-Step Refinement for Symbolic Formulations*

Shashank Kirtania, Priyanshu Gupta, Arjun Radhakrishna
Natural Language Reasoning Workshop at ACL 2024

[NLRSE @ ACL'24]

J.1 *DWT-CompCNN: Deep Image Classification Network for High-Throughput JPEG2000 Compressed Documents*

Tejasvee Bisen, M Javed, Shashank Kirtania, P Nagabhushan
Pattern Analysis and Applications

[Springer 2023]

SELECTED PROJECTS AND COLLABORATIONS

Automated Theorem Proving

Dec. 2024 - Present

Advisor: *Dr. Arun Iyer, Dr. Emily First, Dr. Ashish Tiwari*

MSR India

- Identified **transformer circuits** in foundational models to create activation patches for manipulating formal proof writing. Presented at FM in Wild @ ICLR 2025
- Worked on **formalizing 400+ problems** for INMOBench in lean, and did a empirical study of LLMs and their failures in autoformalization. Oral Presentation at FM for AI workshop @ AAAI
- Working on **Mechanistic interpretability** of LLMs to improve their **ability of mathematical reasoning**.

Automated Code Repair

Jan. 2024 – June 2024

Advisor: *Dr. Gustavo Soares, Dr. Arjun Radhakrishna*

Microsoft PROSE

- Developed **MetaReflection**, an offline reinforcement learning technique **enhancing language agents by leveraging past failures**, improving performance in code **vulnerability detection**. Published @ EMNLP Main.
- This technique showed significant improvements in low resource programming languages and in distilling capabilities of LLMs into SLMs, It is currently **used by GitHub for secret key detection**. deployed by Github Enterprise in 2024
- Worked on **fixing component governance issues from alerts for C#** using tool-aided LLM system. Developed Roslyn APIs as tools to improve code understanding for LLMs.

Knowledge Learning from LLM Interactions

Aug. 2024 – Present

Advisor: *Dr. Arun Iyer, Dr. Gustavo Soares*

MSR India

- Developed **STACKFEED**, a multi-agent actor-critic framework for **iterative knowledge base refinement** using expert feedback, enhancing Retrieval-Augmented Generation (RAG) systems by up to 8% over baselines.
- Built pipeline to use feedback to edits in knowledge bases for **code migration and generation**. Published @ EMNLP 2025
- Worked on learning knowledge bases from interactions of LLM agents Published at MTI @ NeurIPS

Language Models for Automated Reasoning

Apr. 2024 – Present

Advisor: *Dr. Arun Iyer, Dr. Arjun Radhakrishna*

- Introduced **Logic-LM++**, a **multi-step refinement framework** leveraging LLMs for symbolic reasoning, achieving significant improvements over existing methods. Published at NLRSE Workshop @ ACL
- Built pipeline to capture **activation vectors** representing natural language reasoning for LLMs. Used this technique to improve performance on *miniF2F-test* and *PutnamBench*. Published at FM in Wild Workshop @ ICLR.

Automated Repository Code Migration

Sep. 2023 – May 2024

Advisor: *Dr. Gustavo Soares, Dr. Sumit Gulwani*

Microsoft PROSE

- Built pipeline for using LLMs as pseudo-domain experts that learn from failures to detect code vulnerability.
- Employed novel strategy of automatic instruction learning to optimize performance of LLM agent trajectories.

ACHIEVEMENTS AND POSITION OF RESPONSIBILITIES

- Among top 1,200 students selected for **Google Summer of Code 2022** with Numfocus (PyMC).
- Reviewer at AI4Math workshop at ICML 2025 and at DL4Code workshop at NeurIPS 2025 and at ICML 2026.
- Volunteer at PLDI 2024, ICLR 2025.
- Positioned **6th nationally among 5000 + teams** in Amazon Machine Learning challenge 2023.
- **Bronze Medal** in American Express Time Series Forecasting Challenge 2022.
- PyData Delhi talk on “Introduction to Probabilistic Programming”.

PATENTS

1. *Meta-reflection techniques for learning instructions for language agents*

US Patent US12626064B2

[2026-05-12]